

Fig. 5: Monochrome sensor (left) vs colour sensor (right)

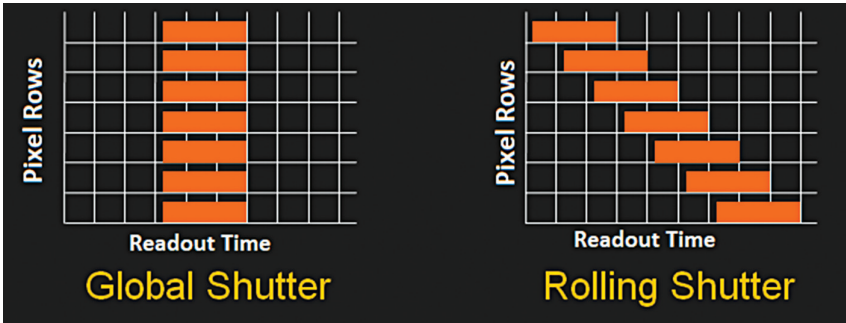


Fig. 6: Pixel row vs time graph of global shutter compared with rolling shutter

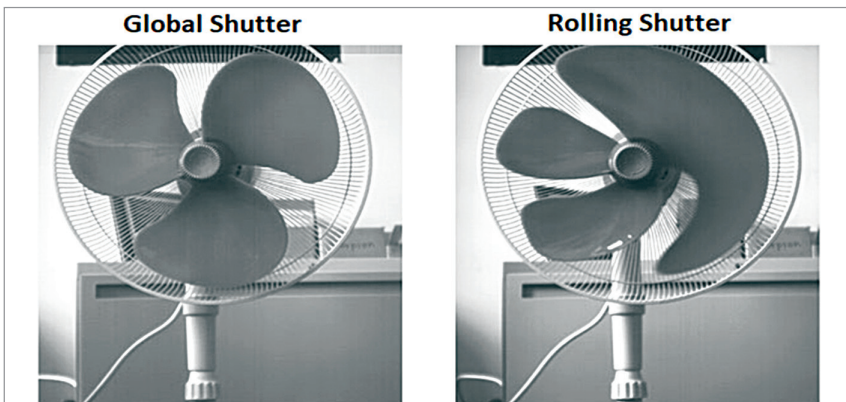


Fig. 7: Comparison of image from camera with a global shutter (left) and rolling shutter (right)
(Source: Link)

sor has an extra layer of colour filter that enables it to capture the colour details. But this filter also reduces the intensity of light falling on the image sensor and thereby slightly reducing the quality of the image. If the application does not require colour detection, a monochrome camera is a better option as it produces better clarity, higher contrast ratio, and does not require any filter.

Global vs rolling shutter. Global shutters were previously available only in CCD sensors, where the whole sensor surface gets exposed to light at the same time, capturing the entire frame at once, which makes it great for high-speed pho-

tography. Rolling shutter exposes the sensor in a progressive way, as shown in Fig. 8, which causes images of fast-moving objects to be distorted.

In rolling shutter technology, each row or column of the image is taken one at a time. The readout process of every row (or column) happens in a few microseconds (μ s), but due to the very high number of frames, the total readout time for the whole image is in the range of a few milliseconds.

Fig. 6 compares the frame time of the rolling shutter with the global shutter. It can be noticed that the complete frame in the rolling shutter

The key considerations while selecting a camera

- Pixel count
- Resolution
- Sensor size
- Sensor technology
- Frame rate
- Chroma
- Shutter speed

is taken over a period of time, whereas the whole frame in the global shutter is taken at the same time.

The difference between rolling shutter and global shutter can be understood from Fig. 7. Applications requiring high dynamic range and low noise should employ CCD sensors, whereas applications with faster speed, low power consumption, and tight budget should go for CMOS sensors.

Frame rate. Frame rate (fps) is the number of images that the sensor can capture and transmit per second. The human brain detects approximately 14 to 16 images per second; the frame rate of a movie is usually 24fps. For fast-moving applications like inspection of newspapers the camera needs to 'shoot' in milliseconds. On the other end there are microscopic applications which require low frame rates comparable to the ones of the human eye.

Recently launched imaging sensors

There have been a lot of technical advancements in imaging technologies, and 2021 has witnessed a lot of imaging sensors developed for smartphones, machine vision, etc. Some of the sensors that have brought something new to the imaging technology are mentioned below.

On Semiconductor XGS 16000. On Semiconductors has introduced the new XGS 16000 to its XGS line-up of CMOS image sensors. It is a 16MP sensor with low power consumption of 1W at 65fps. The